HalalChain

A Simple Guide to Our Research and Application

For reviewers, stakeholders, and non-technical readers

Authors: Muhammad Fatih Maulana, Rio Ferdana Sudrajat, Muhammad Rahardian Baihaqi, Muhammad Fahriza Pratama, Yazid Zinky Arisona, Muhammad Fauzan Lubada
Institution: Universitas Islam Negeri Sunan Gunung Djati Bandung, Indonesia
Full paper: IEEE-format manuscript in `paper/main.pdf`
Live demo: https://web-six-ivory-36.vercel.app/

---

1. Why does this research matter?

Indonesia requires halal labeling for many food products. Certification involves producers, government bodies (BPJPH), and religious auditors (often through MUI). Despite progress, three trust problems remain:

1. **Certificates can be copied.** A paper or PDF label does not prove that *this specific production batch* was actually audited.
2. **Central databases can be changed.** If one administrator controls the records, outsiders cannot independently verify whether history was altered.
3. **Existing digital systems are often too expensive.** Large enterprise blockchain platforms assume big companies with IT teams and complex wallets—not typical UMKM (micro and small) food producers.

HalalChain is our answer: a simple, open system that ties each product **batch** to a permanent, public audit record—while letting ordinary consumers check status with a phone, **without installing a crypto wallet**.

---

2. What is HalalChain in one sentence?

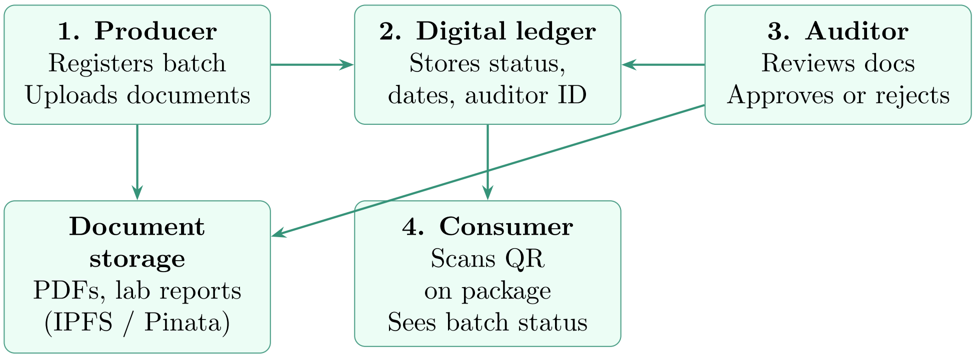
HalalChain is a **batch traceability system** where:

- producers register a batch and upload supporting documents,
- accredited auditors approve or reject that batch on a public digital ledger,
- consumers scan a QR code and instantly see whether that batch is certified, pending, rejected, or revoked.

Think of it as a **transparent audit log** for halal batches—not a replacement for official halal law or religious rulings.

---

3. How does the system work? (Plain-language flow)



What goes where?

- **On the ledger (small, permanent):** batch status, who registered it, who audited it, when, document references, rejection reasons.
- **Off the ledger (large files):** ingredient lists, process photos, lab reports, audit attachments—stored on IPFS (a distributed file network) and linked by a secure reference (CID).

This split keeps costs low while preserving a tamper-evident record of *who decided what, and when*.

---

4. The three user roles in the web application

Role	What they do in the app
Producer	Registers a new product batch, uploads supporting files, receives a batch number and QR code for packaging. Cannot self-certify as halal.
Auditor	Sees pending batches, opens documents, then marks a batch as *Verified*, *Rejected*, or (if needed) *Revoked*. Actions are permanently recorded.
Consumer	Opens a verify link or scans a QR code. Reads the status page only—no account, no wallet, no cryptocurrency needed.

Batch statuses shown to consumers:

- **Verified** — "Tersertifikasi Halal" / Halal Certified
- **Pending** — waiting for auditor review
- **Rejected** — not certified; reason is shown
- **Revoked** — was certified before, but later withdrawn

---

5. Important design rule: rejection cannot be secretly undone

If an auditor **rejects** a batch, that same batch ID can **never** be upgraded to "verified" later. This is enforced by the smart contract, not by policy alone.

If a producer fixes problems, they submit a **new revision batch** linked to the old rejected one. Consumers can still inspect the original rejection and the new approved batch—supporting transparency and accountability (*Amanah*).

---

6. Islamic ethical principles in engineering terms

Our paper maps widely cited Islamic values to concrete system behavior. This is **engineering design**, not a religious fatwa.

Principle	**Meaning**	**How HalalChain implements it**
Amanah	Trustworthiness	Audit outcomes cannot be silently rewritten; rejections stay on record.
Thoyyib	Wholesomeness	The consumer page shows a positive halal result only for verified batches.
Adl	Justice / fairness	Anyone can read batch status from the public ledger; no private gatekeeper API.
Hisba	Accountability	Every approval records *which auditor address* acted and *when*.

---

7. What did we build and test?

Software delivered:

- Smart contract (`HalalChain.sol`) with role-based access control
- Producer, auditor, and consumer web dashboards (Next.js)
- Open-source repository: `https://github.com/Fatihmaull/halal-chain`

Evaluation (Ethereum Sepolia testnet, June 2026):

- **Scenario A:** Batch registered and verified → consumer page shows certified status
- **Scenario B:** Batch rejected, then resubmitted as revision → new batch verified with link to parent
- **Scenario C:** Attempt to approve a rejected batch fails → tamper resistance confirmed
- **Cost signal:** Registering a batch and verifying it cost fractions of a cent in test ETH; reading status for consumers costs nothing

---

8. Try the live prototype

Page	URL
Home	`https://web-six-ivory-36.vercel.app/`
Verified example (Batch #1)	`https://web-six-ivory-36.vercel.app/verify/1`
Rejected example (Batch #2)	`https://web-six-ivory-36.vercel.app/verify/2`
Revision example (Batch #3)	`https://web-six-ivory-36.vercel.app/verify/3`
On-chain contract	`https://sepolia.etherscan.io/address/0xDaCA688e86F438A7cD6B0C9B69606C67CE85Dc92`

---

9. What HalalChain is *not*

- It does **not** replace BPJPH, MUI, or official halal certification law.
- It does **not** automatically analyze ingredients or slaughter methods.
- It does **not** prove halal compliance by itself—it proves **which auditor approved which documents at which time**, on a record that is hard to alter retroactively.

---

10. Research questions addressed

1. Can a **minimal** blockchain design give UMKM-affordable, immutable batch verification?
2. How should we split data between the ledger (status) and file storage (documents) for cost and trust?
3. Can Islamic ethical requirements be translated into **testable** software properties without confusing engineering with religious judgment?

For the full technical treatment—architecture, contract specification, gas tables, limitations, and bibliography—please see the complete paper in `paper/main.pdf`.